A Persistence-Based Invariant for Codimension-Two Bifurcations

Classifying Bogdanov–Takens, cusp, and Bautin points from sampled trajectory data alone, with no access to the governing equations.

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Research summary — interim. This document describes work in progress and is not the research report. It states results obtained to date and the limitations known at the time of writing. A full preprint is in preparation.

Summary. Given only sampled trajectories of a planar dynamical system near a codimension-two bifurcation point — no equations, no parameters, no model — can one determine *which* bifurcation the system is approaching? This project constructs a topological invariant that addresses the question for the three generic planar cases: Bogdanov–Takens, cusp, and Bautin. The construction samples trajectories along an anisotropic path through parameter space, forms a point cloud from their long-time tails, and tracks the resulting persistence barcode as the path angle varies. Three results are reported. A structural one: the H_1 barcode exhibits no discontinuity at a Hopf crossing for any finite integration time, and the reason is critical slowing down rather than a deficiency of the sampling. A quantitative one: despite that, the Hopf angle is recovered to 0.15% of its analytic value by a scaling law, with no parameter tuned toward the known answer. A methodological one: the threshold rule most natural to reach for — a fixed multiple of the median bar length — is disqualified by measurement, because the Betti number it returns scales with sample size.

1 Motivation

A bifurcation is a qualitative change in the long-term behavior of a dynamical system as a parameter varies: an equilibrium loses stability, a periodic orbit is born, two states collide and vanish. Codimension-two points are where two such conditions coincide, and they organize the bifurcation structure of everything nearby. The classical theory of these points is long settled. What is not settled is how to recognize them from *data*.

This matters because in most settings where bifurcations are consequential — ecological collapse, cardiac arrhythmia, power-grid instability, climate tipping — the governing equations are unavailable. What is available is a time series. Existing topological approaches to this problem detect that *some* bifurcation is occurring. They do not say which one. The nearest prior work[1] is single-parameter and codimension-one, and its target is detection rather than classification.

The claim under investigation here is narrower and sharper: the *type* of a codimension-two point is recoverable from trajectory data, and the recovery is by a topological signature rather than by fitting a model.

[1] Tanweer et al., 2024.

2 The construction

2.1 An anisotropic parameter path

The three normal forms studied are

$$\text{BT:}\quad \dot{x} = y,\ \ \dot{y} = \beta_1 + \beta_2 x + x^2 - xy; \qquad \text{cusp:}\quad \dot{x} = \beta_1 + \beta_2 x - x^3;$$

$$\text{Bautin:}\quad \dot{r} = r\,(\beta_1 + \beta_2 r^2 - r^4).$$

Parameter space is sampled along the path $\beta_1 = \varepsilon^4 \cos\theta$, $\beta_2 = \varepsilon^2 \sin\theta$. The anisotropy is not cosmetic. A round circle collapses the quadratic fold and homoclinic curves onto the linear Hopf curve as the radius shrinks, making the crossings numerically unresolvable. With the $(4,2)$ weighting, ε cancels from every crossing condition, so the crossing angles are scale-independent:

Crossing	Analytic angle (rad)
Homoclinic (BT)	4.4831
Hopf (BT)	4.71239
Saddle-node (BT)	4.95071
Fold of cycles (Bautin)	1.80911

2.2 Blow-up coordinates

In raw coordinates the limit cycle near a BT point is a nearly flat sliver that the Vietoris–Rips complex cannot resolve as a loop at all. The rescaling $u = x/\varepsilon^2$, $v = y/\varepsilon^3$ restores it to $O(1)$. The exponents are not fitted: substituting them into the BT normal form and rescaling time by $t = s/\varepsilon$ gives $v' = \cos\theta + u\sin\theta + u^2 - \varepsilon uv$, which is Hamiltonian at $\varepsilon = 0$, with dissipation entering at first order. That balance is what fixes the weights. The Bautin form, whose cycles sit at $r \sim \varepsilon$ in both coordinates, takes isotropic weights $(1,1)$ instead; applying the BT weights to it stretches a round cycle into a thin ellipse and corrupts the H_0 structure.

2.3 Threshold selection

Persistent homology returns a barcode, not a Betti number. Converting one to the other requires a threshold separating genuine features from sampling noise, and that threshold is a modeling decision that has to be justified rather than chosen.

Two rules were implemented and compared. The *median rule* sets the threshold at a fixed multiple of the median bar length. The *gap rule* sorts the bar lengths and cuts at the largest gap among the leading candidates. On a cloud consisting of two well-separated components the two agree, and either is defensible. On the clouds that actually arise here they do not:

Cloud size N	300	600	1200	2400
Median-rule β_0	54	116	249	495
Gap-rule β_0	2	2	2	2

Doubling the sample doubles the median-rule answer. A Betti number is a property of the underlying space and cannot depend on how densely that space was sampled, so the median rule is measuring sampling density rather than topology.

The mechanism was diagnosed rather than assumed, and it contradicted the initial hypothesis. The expectation was a bimodal H_0 death-time distribution — short distances within a trajectory, longer ones between trajectories. The measured distribution is smooth and unimodal with no gap anywhere: the retained tails fill the region near-uniformly, so H_0 deaths are simply nearest-neighbour distances of a roughly uniform planar sample, whose right tail is long. A threshold at twice the median therefore lands inside the bulk — at the 79th percentile in the measured case — and retains a fixed *fraction* of bars, which is why the count tracks N. Recovering $\beta_0 = 2$ would require a multiple of $c \approx 11.3$, and that value is cloud-specific, so no fixed multiple works across angles. The gap rule is used because it responds to the shape of the persistence distribution rather than its absolute scale.

2.4 Undefined is not zero

Beyond the fold curve no bounded attractor exists: trajectories escape and there is nothing to sample. The invariant is reported as UNDEFINED there, never as $\beta_0 = \beta_1 = 0$. This is a domain failure, not a measurement, and conflating the two would manufacture a false discontinuity at the fold.

2.5 Backward-time union

The inner limit cycle of the Bautin unfolding is repelling and therefore invisible to forward integration at any resolution. The cloud is formed as the union of forward and backward trajectory tails, with the domain hypothesis stated explicitly as part of the construction rather than left in a code comment.

3 Results

Saddle-node: detection by basin extinction. The count of bounded trajectories reaches zero at $\theta = 4.95076$, against the analytic value 4.9507064 — one grid step. The detector here is not topological: the saddle that annihilates is never an attractor and never appears in trajectory-sampled data at all. What is observed is the extinction of the basin. This is reported as such, and it is the reason the invariant is stated as a hybrid of Betti numbers and a bounded-trajectory count rather than as Betti numbers alone.

Hopf: a structural negative, and a recovery. No Betti discontinuity appears at the Hopf crossing — at any threshold, any transient truncation, or any integration time tested. Truncating the transient $30\times$ harder changes the longest H_1 bar by a median of 6%. The reason is structural. Fitting the measured persistence against the linear decay rate gives

$$\log(\text{persistence}) \;=\; 0.998\left(\operatorname{Re}\lambda \cdot t_{\text{end}}\right) + \text{const}, \qquad R^2 = 0.9993,$$

a coefficient of unity meaning the residual loop is exactly $\exp(\operatorname{Re}\lambda\, t_{\text{end}})$: a spiral that has not finished converging. The required convergence time $1/|\operatorname{Re}\lambda|$ diverges as the Hopf point is approached, so no feasible integration time removes it. This is critical slowing down, which is classical; what is new here is its consequence for trajectory-union persistent homology.

The bifurcation is nonetheless present in the data, encoded in the *shape* of the decay rather than as a discontinuity. A supercritical Hopf gives amplitude $\sim \sqrt{\theta_H - \theta}$, so the squared persistence should be linear below the crossing and vanish at it. It is ($R^2 = 0.992$–0.997). Across seven fit windows chosen without reference to the Hopf angle, the extrapolated zero lands in $[4.7084, 4.7194]$, against the analytic 4.71239 — within ± 0.007 rad, or 0.15%, with nothing tuned toward the answer.

An earlier version of this fit used the Hopf angle itself as the right edge of the fit window, which made the extrapolation circular and produced a falsely tight ± 0.0007 rad. That figure is withdrawn; the honest interval is the one above.

Bautin: the repelling cycle. With the backward-time union, $\beta_1 = 2$ is detected on the contiguous block $\theta \in [1.7065, 1.7848]$, which lies inside the analytic two-cycle window $(\pi/2, 1.8091)$. Forward integration alone gives $\beta_1 = 1$ throughout. Detection is partial within the window, as expected: the inner cycle radius vanishes at the lower edge and is below sampling resolution there.

4 Errors found and corrected

Three are reported because the diagnosis is part of the result.

At the cusp's Hopf angle every trajectory collapses to a point, leaving a cloud of diameter 1.8×10^{-8} — solver roundoff — from which the persistence routine reported a spurious loop. A degeneracy guard now returns $\beta_0 = 1$, $\beta_1 = 0$ at such angles: a point attractor is a bounded object with no holes, which is a measured answer and not an absent one.

Backward-time clouds assembled from as few as four trajectories formed thin arcs that registered as loops corresponding to no invariant set. A minimum-bounded-trajectory guard removes these; the genuine detection survives it.

Separately, filtering to the densest 80% of a cloud at $\theta = 4.65$ raised the H_1 persistence from 0.514 to 0.770, indicating that trajectories crossing the interior of the cycle depress the measured persistence by roughly half. This is recorded as a limitation, not corrected.

5 Status and limitations

The classical unfoldings are derived by hand and complete. The computational pipeline is built, instrumented, and under version control, with every run logging its full parameter set. The results above are reproducible from that pipeline.

What is not established: a formal proof of the correspondence between barcode discontinuities and bifurcation crossings; the classification result for all three types stated as a theorem rather than demonstrated on normal forms; and validation on a system that was not constructed as a normal form. The intended validation target is Bazykin's predator–prey model, which has a genuine Bogdanov–Takens point and a known analysis to check against.

Two scope limits are worth stating plainly. The negative result concerns *thresholded* Betti numbers under the gap rule, not the barcode itself — the stability theorem guarantees the barcode varies continuously wherever the cloud does, so no claim is made against it. And the invariant as constructed depends on the sampling protocol; whether the signature is invariant across protocols is an open question currently under investigation, and the classification claim is provisional until it is settled.

Preprint in preparation. Correspondence: Team Arenda.